Evaluating Agents on Scientific Discovery
A Benchmark with Provable Ground Truth from Arithmetic Statistics

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Motivation

Evaluation of LLM-based agents on scientific discovery is methodologically circular: scoring a discovered law presupposes knowledge of the true law, which is unavailable precisely for the problems of interest. Current discovery benchmarks substitute LLM judges or rubric matching, inheriting the reliability problems of the systems under evaluation. A second confound is contamination: when a system states a correct law, standard evaluations cannot distinguish derivation from memorization of published results.

We address both issues by construction. The benchmark plants laws that are provable theorems, drawn from arithmetic statistics, so that discovery can be scored exactly against a known answer key, and generates task variants absent from the literature, so that memorization and reasoning are separately identifiable.

Task family

Let $K/\mathbb{Q}$ be a Galois number field with $G = \mathrm{Gal}(K/\mathbb{Q})$, and let $\zeta_K(s) = \sum_{n\geq 1} a_n(K)\, n^{-s}$ be its Dedekind zeta function, so that $a_n(K)$ counts ideals of norm n. For a rational prime p with decomposition type (e, f, g) in K (ramification index, inertia degree, and number of primes above p, with $efg = [K{:}\mathbb{Q}]$), the p-Euler factor is $(1 - p^{-fs})^{-g}$, whence

$$a_{p^k}(K) \;=\; \binom{g + k/f - 1}{g - 1} \text{ if } f \mid k, \qquad a_{p^k}(K) = 0 \text{ otherwise,}$$

and a_n is determined multiplicatively. The inertia degree f at an unramified prime is the order of its Frobenius class, which by Chebotarev is equidistributed with respect to the counting measure on G. Consequently the isomorphism type of G leaves provable statistical fingerprints in the coefficient vector $(a_1, \ldots, a_N)$: only element orders of G occur as inertia degrees, with prescribed densities. A representative criterion: for $[K{:}\mathbb{Q}] = \ell^2$, the extension is cyclic iff $a_{p^\ell}(K) = 0$ for some prime p; only C_{ℓ^2} admits inert primes, and in the noncyclic case a_{p^ℓ} is provably nonvanishing at every prime, ramified or not [1]. Each benchmark task is a classification problem: given $(a_1, \ldots, a_N)$ for a field drawn from this generative model, identify G among the groups of the given order, and state the distinguishing criterion. The setting follows Lee and Lee [1], who classified such G with interpretable models and promoted the learned decision logic to new theorems.

The benchmark distribution is the exact Chebotarev model: independent Frobenius classes at each prime, i.e. the idealized unramified statistics. This idealization is validated against genuine arithmetic: on number fields drawn from the LMFDB (with zeta data computed from defining polynomials via Dedekind's theorem), the proven quartic criterion classifies at accuracy 1.000, as the theorem guarantees.

Exact ground truth

Because the generative model is known in closed form, each task ships with quantities that are typically unobtainable:

- **Bayes-optimal accuracy.** The class-conditional law of the observable prime signatures is an explicit categorical distribution per prime, so the Bayes risk is estimable to $\pm 5 \times 10^{-4}$ by direct signature sampling.

Agent performance is reported as regret against this ceiling; the ceiling itself calibrates task difficulty (from 0.9998 down to 0.84 across the suite).

- **Exact attribution ground truth.** The mutual information $I(a_n; G)$ is computable in closed form for every feature via the independence of prime-power factors. A system's stated feature attributions are scored against this ranking (rank correlation, precision at k), which exposes cases where predictive accuracy and explanation fidelity decouple; we observe decision trees within 0.3% of the ceiling whose importances correlate weakly with the true ranking.
- **Reference theorems.** The correct explanation for each task is a proven statement (e.g. the inert-prime criterion above), serving as the answer key for conjecture grading.

A natural hardening step is formal verification of the answer keys. The discrete content of each criterion (the element-order multisets of the candidate groups, the vanishing pattern of the Euler coefficients as a function of the inertia degree, and the resulting exclusion arguments) is directly formalizable in Lean 4 against Mathlib, making the ground truth machine-checked rather than merely peer-reviewed. The analytic inputs (Euler products, the Chebotarev density theorem) would enter as stated hypotheses pending their availability in Mathlib. We view machine-checked answer keys as the natural endpoint of the benchmark's premise: an evaluation is only as trustworthy as its ground truth.

Evaluation protocols and contamination design

Systems are evaluated under three protocols: *rule induction* (data and mathematical context provided; the system must return an executable decision rule and a stated conjecture, with the rule scored on held-out fields), *derivation* (mathematical context only, no data; tests whether the criterion can be derived rather than fit), and *direct classification* (few-shot in-context prediction). The harness is model-agnostic: any prompt-to-text interface, including a multi-agent system behind a single entry point, plugs in unchanged; alternatively an agent may be given the dataset files and evaluated on its returned artifacts. Sandboxed execution and oracle/majority controls validate the pipeline independently of any model.

The generator requires only the element-order multiset of a candidate group, so the suite includes tasks for degrees 14, 21, and 25 whose distinguishing criteria follow from the same theory but, to our knowledge, appear in no publication. Matched comparisons between published and held-out tasks convert contamination from a hypothesis into a measurement.

Preliminary findings

A preliminary single-seed run of Claude Sonnet 4.5 exhibits the separations the design targets. Under rule induction the model attains near-ceiling regret on several tasks, including held-out degree 14 (regret 0.007). Under derivation it collapses to majority-class performance on most tasks, with one exception: the degree-9 task, whose criterion is published, where it achieves regret 0.004. This asymmetry is consistent with memorization of the published criterion; the decisive comparison is against the structurally matched held-out degree-25 task (both are the ℓ^2 family), which is part of the full multi-seed evaluation in progress. Notably, the model also scored 0.958 on the held-out degree-14 derivation, where no published criterion exists; whether its stated conjecture constitutes a correct derivation of the inert-prime argument is exactly what the conjecture-grading protocol assesses, and the harness now records all model outputs for this analysis.

Directions: a testbed for trustworthy multi-agent ML

Beyond capability evaluation, the framework instruments security and trust questions that are normally unmeasurable because harm cannot be quantified without ground truth.

Poisoning with exact damage accounting. Since the generative model is known in closed form, training-distribution attacks can be analyzed information-theoretically: perturbing the Chebotarev distribution shifts the Bayes ceiling by a computable amount, separating the information the attack destroys from the performance the learner loses. Standard poisoning studies cannot make this distinction.

Byzantine tolerance of agent pipelines. In a multi-agent discovery system, a compromised member can inject subtly flawed analyses. With exact regret as the harm metric, degradation can be plotted against the number and subtlety of compromised agents, and the detection value of verifier roles measured directly. This is Byzantine fault tolerance for epistemic outputs rather than messages.

Faithfulness auditing. Gradeable explanations expose systems that are right for the wrong reasons; we already observe models within 0.3% of the ceiling whose stated feature attributions correlate weakly with the true information ranking. In a security setting this is the difference between passing accuracy gates and being trustworthy.

A first instrument for these questions is implemented: a tool-using multi-agent pipeline (Strands agents on Amazon Bedrock) in which an analyst agent explores the data and proposes a rule, and an independent critic agent verifies or revises it. Four conditions (analyst alone, analyst with critic, a simulated compromised analyst, and the compromised analyst with critic) make the value of verification and the harm of a compromised member directly comparable on the same tasks; this evaluation is in progress. These questions seem complementary to the APWA evaluation agenda.

Code, benchmark suite (18 configurations, Hugging Face format), and harness: `https://github.com/mmjerge/galois-ml`.

References

[1] K.-H. Lee and S. Lee, *Machines Learn Number Fields, But How? The Case of Galois Groups*, arXiv:2508.06670 [math.NT], 2025. `https://arxiv.org/abs/2508.06670`